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9th Grade Chemistry · Non-AP · Ala Vistancia · Mr/Ms Sharp

Foundations of Chemistry — Answer Key

SECTION A: VOCAB (17) · SECTION B: MCQ (20) · SECTION C: FIB (13) · TOTAL: 50 PTS

QUICK REFERENCE

All Answers At a Glance

SECTION A · VOCABULARY (Q1–Q17)

Q1	C.	atom
Q2	B.	element
Q3	E.	compound
Q4	D.	mixture
Q5	E.	proton
Q6	F.	neutron
Q7	E.	electron
Q8	E.	atomic number
Q9	F.	mass number
Q10	E.	isotopes
Q11	F.	ion
Q12	D.	valence
Q13	E.	covalent
Q14	F.	ionic
Q15	C.	mole
Q16	E.	reactant
Q17	F.	product

SECTION B · MULTIPLE CHOICE (Q18–Q37)

Q18	D.	Ice melting
Q19	C.	Homogeneous mixture
Q20	E.	9.0 g/cm^3 ★
Q21	E.	In the nucleus
Q22	D.	35 ★
Q23	C.	Proton
Q24	F.	Chemical properties
Q25	F.	Argon (Ar)
Q26	E.	Atomic number
Q27	F.	Ionic bond
Q28	E.	4 valence e^-
Q29	C.	Bent (V-shaped)
Q30	F.	Mass and atoms
Q31	F.	$2H_2O_2 \rightarrow 2H_2O+O_2$
Q32	E.	Moles/molecules
Q33	D.	1 mole
Q34	F.	18.0 g/mol ★
Q35	F.	2.0 mol ★
Q36	E.	Liquid
Q37	F.	Increases

SECTION C · FILL-IN-BLANK (Q38–Q50)

Q38	C.	protons
Q39	D.	ion
Q40	C.	isotopes
Q41	D.	covalent
Q42	D.	group
Q43	E.	reactants
Q44	E.	coefficients
Q45	E.	physical
Q46	E.	nucleus
Q47	C.	mole
Q48	D.	44 g/mol ★
Q49	E.	aqueous
Q50	E.	metals

★ = calculation model provided

SECTION A · 17 POINTS · 1 POINT EACH

Vocabulary — Answer Key

ATOM
Smallest particle of an element retaining its chemical properties.

ELEMENT
Pure substance of one type of atom; cannot be chemically broken down further.

COMPOUND
Pure substance of 2+ elements chemically bonded in a fixed ratio.

MIXTURE
2+ substances NOT chemically combined; each retains own properties.

PROTON
Positive (+) subatomic particle in the nucleus. Count = atomic number.

NEUTRON
Neutral subatomic particle in the nucleus. Changing neutrons → isotopes.

ELECTRON
Negative (−) subatomic particle in orbitals around the nucleus.

ATOMIC NUMBER
Number of protons in the nucleus. Unique to each element; never changes.

MASS NUMBER
Protons + Neutrons in the nucleus.

ISOTOPE
Same element (same protons), different neutron count, different mass number.

ION
Atom that gained electrons (anion, −) or lost electrons (cation, +).

VALENCE ELECTRONS
Outermost electrons; responsible for chemical bonding.

COVALENT BOND
Bond formed by SHARING electrons; usually between nonmetals.

IONIC BOND
Bond formed by TRANSFERRING electrons; metal to nonmetal.

MOLE

REACTANT

SI unit for amount; 1 mol = 6.022×10^{23} particles (Avogadro's number).

Starting substance consumed in a reaction; LEFT side of equation.

PRODUCT

New substance formed in a reaction; RIGHT side of equation.

SECTION B · 20 POINTS · 1 POINT EACH

Multiple Choice — Answer Key with Rationale

L01 · MATTER & MEASUREMENT

Q18 · Physical vs. Chemical Change

D. Ice melting

Which of the following is an example of a physical change?

CORRECT ANSWER

D. Ice melting into liquid water

Physical change: only the state of matter changes (solid → liquid). H_2O remains H_2O. No new substances are formed. Rusting (A), burning (B), curdling (C), rotting (E), and the baking soda reaction (F) all produce new chemical substances — those are chemical changes.

Q19 · Classification of Matter

C. Homogeneous mixture

Salt water (NaCl dissolved in H_2O) is best classified as a...

CORRECT ANSWER

C. Homogeneous mixture

Salt water contains two substances not chemically combined (NaCl + H_2O). The salt can be recovered by evaporation. Composition is uniform throughout (homogeneous). It is NOT a pure compound because the ratio of salt to water can vary.

Q20 · Density Calculation ★

E. 9.0 g/cm³

A metal sample: mass = 45.0 g, volume = 5.0 cm^3. Density?

DENSITY CALCULATION — FULL WORK

Formula: D = m ÷ v

Values: m = 45.0 g | v = 5.0 cm³

Solve: D = 45.0 ÷ 5.0

Answer: D = 9.0 g/cm³

Common errors: dividing volume by mass (gives 0.11) or multiplying (gives 225). Always identify the formula first: D = m/v.

L02 · ATOMIC STRUCTURE

Q21 · Location of Atomic Mass

E. In the nucleus

Where is the majority of an atom's mass located?

CORRECT ANSWER

E. In the nucleus

Protons (~1 amu each) and neutrons (~1 amu each) are both in the nucleus. Electrons are ~1/1836 the mass of a proton — negligible. Rutherford's gold-foil experiment confirmed that nearly all mass is concentrated in a tiny, dense nucleus.

Q22 · Mass Number Calculation ★

D. 35

An atom has 17 protons and 18 neutrons. Mass number?

MASS NUMBER CALCULATION — FULL WORK

Formula: Mass Number = Protons + Neutrons

Values: Protons = 17 | Neutrons = 18

Solve: 17 + 18

Answer: Mass Number = 35 (Chlorine-35, ^{35}Cl)

Note: atomic number (17) = element identity (Chlorine). Mass number (35) = this specific isotope.

Q23 · Element Identity

C. Proton

Which subatomic particle determines the identity of an atom?

CORRECT ANSWER

C. Proton (via atomic number)

Change proton count → different element. Change neutrons → isotope of the SAME element. Change electrons → ion of the SAME element. The atomic number (proton count) is the defining characteristic.

L03 · PERIODIC TABLE

Q24 · Groups vs. Periods

F. Chemical properties

Elements in the same group share similar...

CORRECT ANSWER

F. Chemical properties

Same group = same number of valence electrons = similar chemical behavior. Example: Group 1 (Li, Na, K) all have 1 valence electron and all react vigorously with water. Periods share the same number of electron shells, not valence electrons.

Q25 · Noble Gases

F. Argon (Ar)

Which element is a noble gas?

CORRECT ANSWER

F. Argon (Ar) — Group 18

Noble gases (Group 18): He, Ne, Ar, Kr, Xe, Rn. Full valence shells = chemically stable/unreactive. Distractors: N (Group 15), O (Group 16), F (Group 17/halogen), Cl (Group 17/halogen), Na (Group 1/alkali metal).

Q26 · Periodic Trends

E. Atomic number

Moving left to right across Period 3, which property steadily increases?

CORRECT ANSWER

E. Atomic number

Atomic number increases by exactly 1 for each element across any period — that is how the table is organized. Atomic radius DECREASES across a period (more protons pull electrons closer). Metallic character also DECREASES moving right.

L04 · CHEMICAL BONDING

Q27 · Ionic vs. Covalent

F. Ionic bond

Which bond involves the complete TRANSFER of electrons?

CORRECT ANSWER

F. Ionic bond

Ionic bond: electrons fully transferred from metal (loses e^- → cation) to nonmetal (gains e^- → anion). Na^+ and Cl^- attract and form NaCl. Covalent bond: electrons are SHARED (not transferred).

Q28 · Valence Electrons of Carbon

E. 4

How many valence electrons does carbon (Group 14) have?

CORRECT ANSWER

E. 4 valence electrons

For main-group elements, the group number gives the valence electron count: Group 14 = 4 valence electrons. Carbon can form 4 bonds — fundamental to all organic chemistry and life.

Q29 · Molecular Shape of Water

C. Bent (V-shaped)

H_2O has two lone pairs on O. What molecular shape results?

CORRECT ANSWER

C. Bent (V-shaped) — ~104.5° bond angle

Oxygen has 4 electron groups (2 bonds + 2 lone pairs) → tetrahedral electron geometry. Molecular shape only counts ATOMS, giving bent. The lone pairs push H atoms closer together. This bent shape + electronegativity difference = polar molecule = why water has surface tension, high boiling point, etc.

L05 · CHEMICAL REACTIONS

Q30 · Law of Conservation of Mass

F. Mass & number of atoms

What is always conserved in a chemical reaction?

CORRECT ANSWER

F. Total mass and total number of each type of atom

This is the Law of Conservation of Mass (Lavoisier, 1789). Matter is neither created nor destroyed. This is exactly why we balance chemical equations — the number and type of atoms on the left must equal the right.

Q31 · Reaction Types

F. $2H_2O_2 \rightarrow 2H_2O + O_2$

Which represents a decomposition reaction?

CORRECT ANSWER

F. $2H_2O_2 \rightarrow 2H_2O + O_2$ (hydrogen peroxide decomposing)

Decomposition: ONE compound → TWO or more products. Reaction type reference: A = synthesis, B = combustion, C = single-replacement, D = double-replacement (neutralization), E = synthesis.

Q32 · Coefficients vs. Subscripts

E. Moles / molecules

What do coefficients in a balanced equation represent?

CORRECT ANSWER

E. The number of moles (or molecules) of each substance

COEFFICIENT (large, in front): number of moles/molecules. SUBSCRIPT (small, inside formula): number of atoms in one molecule. NEVER change subscripts to balance — that changes the substance itself (e.g., $H_2O \rightarrow H_3O$ changes the compound).

L06 · STOICHIOMETRY

Q33 · Avogadro's Number

D. 1 mole of any substance

6.022×10^{23} represents the number of particles in exactly...

CORRECT ANSWER

D. 1 mole of any substance

By definition: 1 mole = 6.022×10^{23} representative particles. This applies to atoms, molecules, ions, formula units, etc. Avogadro's number was chosen so that atomic mass in amu = molar mass in g/mol.

Q34 · Molar Mass of H_2O ★

F. 18.0 g/mol

Molar mass of H_2O? (H = 1.0 g/mol, O = 16.0 g/mol)

MOLAR MASS — H_2O — FULL WORK

Formula: MM = Σ (# atoms × atomic mass)

Hydrogen: 2 atoms × 1.0 g/mol = 2.0 g/mol

Oxygen: 1 atom × 16.0 g/mol = 16.0 g/mol

Total MM = 18.0 g/mol

Common error: students add all atomic masses without multiplying by count. 2 H atoms × 1.0 = 2.0, NOT just 1.0.

Q35 · Moles from Mass ★

F. 2.0 mol

How many moles in 36.0 g of water? (MM = 18.0 g/mol)

MOLES CALCULATION — FULL WORK

Formula: n = m ÷ M

n = moles | m = mass (g) | M = molar mass (g/mol)

Values: m = 36.0 g | M = 18.0 g/mol

Solve: n = 36.0 ÷ 18.0

Answer: n = 2.0 mol

Intuition check: if 1 mol = 18 g, then 36 g = 2 mol. Like: if 1 dozen = 12 eggs, 24 eggs = 2 dozen.

L07 · STATES OF MATTER & SOLUTIONS

Q36 · States of Matter

E. Liquid

Which state has a definite volume but no definite shape?

CORRECT ANSWER

E. Liquid

Solid: definite shape + definite volume (particles locked). Liquid: definite volume + no definite shape (particles close but can slide past each other). Gas: no definite shape + no definite volume (particles spread to fill container).

Q37 · Solubility & Temperature

F. Increases

As temperature increases, solubility of a solid solute in water...

CORRECT ANSWER

F. Increases (for most solid solutes)

Higher temperature → faster-moving water molecules → can pull more solid solute into solution. IMPORTANT EXCEPTION to mention: solubility of gases in liquids DECREASES with temperature (warm soda goes flat faster).

SECTION C · 13 POINTS · 1 POINT EACH

Fill-in-the-Blank — Answer Key

Q38 · Atomic Structure

CORRECT ANSWER

C. protons

Proton count = atomic number = element identity.

Q39 · Ions

CORRECT ANSWER

D. ion

Gains electrons → anion (−). Loses electrons → cation (+).

Q40 · Isotopes

CORRECT ANSWER

C. isotopes

Same element, same protons, different neutrons.

Q41 · Covalent Bond

CORRECT ANSWER

D. covalent

Sharing = covalent. Transferring = ionic.

Q42 · Groups

CORRECT ANSWER

D. group

Group = column. Period = row.

Q43 · Reactants

CORRECT ANSWER

E. reactants

Left of arrow = reactants. Right = products.

Q44 · Coefficients

CORRECT ANSWER

E. coefficients

Never change subscripts when balancing.

Q45 · Physical Change

CORRECT ANSWER

E. physical

No new substance formed; salt recoverable by evaporation.

Q46 · Nucleus

CORRECT ANSWER

E. nucleus

Contains protons + neutrons; holds nearly all atomic mass.

Q47 · The Mole

CORRECT ANSWER

C. mole

1 mole = 6.022×10^{23} particles (Avogadro's number).

Q48 · Molar Mass of CO_2 ★

MOLAR MASS — CO_2 — FULL WORK

Carbon: 1 atom × 12 g/mol = 12 g/mol

Oxygen: 2 atoms × 16 g/mol = 32 g/mol

Total MM = 44 g/mol → Answer: D. 44

Q49 · Aqueous Solution

CORRECT ANSWER

E. aqueous

Aqueous = water as the solvent (Latin: aqua).

Q50 · Metals

CORRECT ANSWER

E. metals

Metals: shiny, malleable, ductile, good conductors (~75% of all elements).

SCORING GUIDE

Grade Scale Reference

90–100 45–50 pts · A	**80–89** 40–44 pts · B	**70–79** 35–39 pts · C	**60–69** 30–34 pts · D
Below 60 Below 30 pts · F			